VehicleSure: Enterprise Vehicle Insurance & Claims Management System

Automating risk evaluation, underwriting decisions, and claim settlement for modern insurance operations

Developer: Hari Charan Lekkala

Tech Stack: Spring Boot · Angular SPA · JWT Auth · RBAC · MySQL · Tailwind CSS

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The Broken State of Traditional Vehicle Insurance

Slow Manual Processes

Manual underwriting leads to inconsistent decisions and delayed policy approvals

Paper-Based Filing

Paper-based claim submission causes significant settlement delays and errors

Disconnected Workflows

No unified system between customers, underwriters, and claims officers

Fraud Vulnerability

High risk of fraudulent claims slipping through inadequate verification

Zero Transparency

Poor customer experience with no visibility on application or claim status

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VehicleSure: One Platform. Every Stakeholder.

A comprehensive digital insurance ecosystem that connects all stakeholders through role-based portals, delivering automated premium calculation, intelligent risk assessment, and real-time application tracking.

Automated Premium Calculation

Instant quotes based on vehicle details and risk profiles

Intelligent Risk Assessment

Data-driven underwriting decisions with consistent criteria

Real-Time Tracking

Live visibility on application and claim status for all users

Digital Document Generation

Automated policy and invoice PDF creation with OpenPDF

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Key Terminology

Understanding essential terms is crucial for navigating the VehicleSure platform and the vehicle insurance landscape.

Term	Definition
IDV (Insured Declared Value)	Current market value after depreciation; the maximum financial exposure
Third Party (TP)	Mandatory liability coverage adjusted by vehicle category (e.g., Heavy or Light)
Own Damage (OD)	Coverage for the insured vehicle, influenced by loading factors
Comprehensive Total Loss (GTL)	Condition where repair costs exceed 75% of IDV

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Use Case 1: Policy Purchase Workflow

01

User Applies for Policy

Customer submits application with vehicle details

02

Admin Assigns Underwriter

Admin reviews and assigns to available underwriter

03

Underwriter Reviews & Decides

Underwriter accepts or rejects the application

04

If Accepted: User Pays Premium

Customer completes payment to buy the policy

05

Policy Active

Policy is activated and coverage begins

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Use Case 2: Claims Processing Workflow

01

User Files Claim

Customer submits claim with incident details and proof documents

02

Admin Assigns Claims Officer

Admin reviews and assigns to available claims officer

03

Claims Officer Reviews & Decides

Claims officer accepts or rejects the claim

04

If Accepted: Admin Processes Payment

Admin verifies and initiates payout to customer

05

Claim Settled

Claim is closed and payment is completed

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From Quote to Submission in Minutes

01

Vehicle Details Entry

Customer enters ex-showroom price, age, mileage, and accident history

02

IDV Calculation

System calculates Insured Declared Value using formula: Ex-Showroom Price × (1 − 0.10 × Vehicle Age), capped at 50% depreciation

03

Premium Display

Instant premium shown based on risk level and IDV bracket

04

Document Upload

Customer uploads Registration Certificate and invoice documents

05

Application Submitted

Status updates to **SUBMITTED → UNDER REVIEW** in real-time

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Intelligent Risk Scoring at the Core

Risk Assessment Matrix

Factor	Criteria	Risk Level
Vehicle Age	>5 years	High
Accident History	Multiple claims	High
Total Mileage	>100,000 km	Medium/High

Premium Formula

Final Premium = Base Premium × Risk Multiplier × IDV Multiplier × Tenure Discount

1.0x
Low Risk

1.2x
Medium Risk

1.5x
High Risk

IDV Multipliers

>10L: +30% | 5–10L: +10% | <2L: −10%

Tenure Discounts

2-year policy: 1.9x | 3-year: 2.8x

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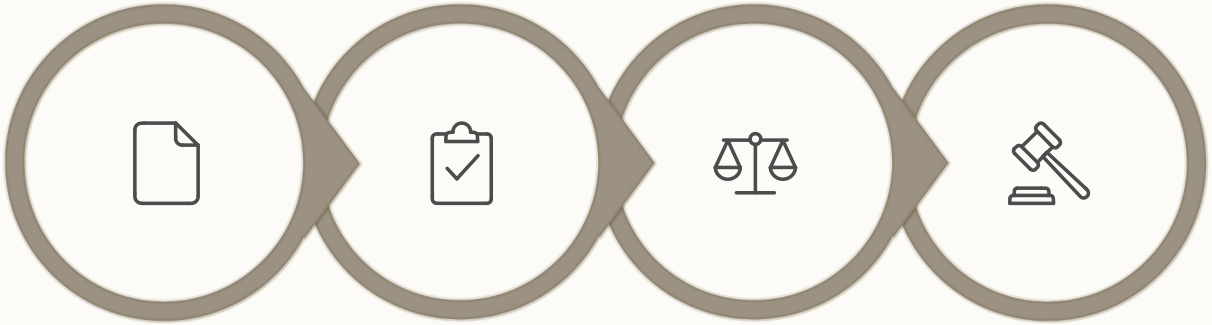
End-to-End Claim Lifecycle Management



File Claim

Review Claim

Calibrate Amount

Final Decision

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Future Scope & Technology Stack

Future Enhancements

AI-Driven Risk Scoring

Machine learning models for predictive underwriting

OCR Document Verification

Automated extraction from Registration Certificate and invoices

Advanced Fraud Detection

Anomaly detection algorithms for suspicious patterns

Mobile Applications

Dedicated apps for customers and insurance staff

Database Integration

Direct RTO and vehicle history database connections

Predictive Analytics

Dashboard for underwriter decision support

Technology Stack

- Java 17
- Spring Boot 3.3
- Angular
- Single Page App (SPA)
- Role-Based Access Control (RBAC)
- RxJS Signals
- MySQL
- JWT Security
- OpenPDF
- Tailwind CSS
- Swagger/OpenAPI

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